Q.1 Which of the following queue does not always follow FIFO behavior?

A. Simple queue
B. Circular queue
C. Double-ended queue
D. Priority queue

Q.2 The depth of a complete binary tree is 6 , the number of its non-leaf nodes will be:

A. 6
B. 61
C. 64
D. 63

Q.3 The _____ level of any leaf in a binary tree is called the depth of the tree

A. Maximum
B. Minimum
C. Mid
D. Zero

Q.4 A binary tree in which every non-leaf node has non-empty right and left subtrees is called:

A. AVL Tree
B. Strictly Binary Tree
C. Graph
D. Huffman Tree

Q.5 If there is a complete binary tree of depth 4, the total number of nodes in it will be:

A. 30
B. 33
C. 31
D. 32

Q.6 In a table, column represents _____.

A. Data
B. Attributes
C. Instances
D. Records

Q.7 Table is a _____ representation of a relation.

A. Two Dimensional
B. Three Dimensional
C. Four Dimensional
D. Six Dimensional

Q.8 Consider the following two sets: STD Name = [Ali, Hassan] Course Name = {Eng, Math, Phy} Then the Cartesian product of STD Name and Course Name (i.e. STD Name x Course Name) will be equal to:

A. {(Ali, Math), (Ali, Phy), (Hassan, Phy), (Hassan, Eng), (Hassan, Math), (Hassan, Phy)}
B. {(Ali, Eng), (Hassan, Math), (Ali, Phy), (Hassan, Eng), (Hassan, Math), (Hassan, Phy)}
C. {(Ali, Eng), (Ali, Math), (Ali, Phy), (Hassan, Phy), (Ali, Math), (Hassan, Eng)}
D. {(Ali, Eng), (Ali, Math), (Ali, Phy), (Hassan, Eng), (Hassan, Math), (Hassan, Phy)}

Q.9 Consider the following two sets: X = [a, b] Y = {1,3,5} Then the Cartesian product of A and B (i.e. XxY) will be equal to:

A. {(a,1), (a,5), (a,3), (b,5), (b,1), (b,5)}
B. {(a, a), (a,3), (a,1), (b,5), (b,5), (b,3)}
C. {(a,1), (b,1), (a,5), (b,3), (a,5), (b,5)}
D. {(a,1), (a,3), (a,5), (b,1), (b,3), (b,5)}

Q.10 _____ represents characteristics of an object.

A. Modality
B. Attribute
C. Relationship
D. Cardinality

Q.11 The instances of _____ cannot exist independently?

A. Hybrid Entity
B. Strong Entity
C. Weak Entity
D. Naming Entity

Q.12 A cell of a table is the intersection of row and a _____.

A. Tuple
B. Degree
C. Column
D. Record

Q.13 According to the definition of an entity, which of the following is NOT included in it?

A. Concept
B. Person
C. Action
D. Object

Q.14 In relational database, each cell of a table cannot have _____ .

A. Unique values
B. String values
C. Numeric values
D. Multiple values

Q.15 The transfer of characteristics of super type to sub types is knows as _____.

A. Abstraction
B. Inheritance
C. Dependency
D. Encapsulation
